

AYDIN EZZATPOUR AGHDAM

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EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. in Electrical Engineering; GPA: 18.46/20

2023 - Present

- **Selected coursework:** Engineering Mathematics (20/20), Digital Signal Processing (20/20), Convex Optimization (19.6/20), Introduction to Machine Learning (19.4/20), and Linear Algebra (19/20); currently taking graduate courses in Information and Coding Theory and Probabilistic Analysis in High Dimensions.
- **Honor:** Ranked 119th in Region 1 in the Iranian Nationwide University Entrance Examination (2023).

RESEARCH INTERESTS

Federated Learning; Information-Theoretic Machine Learning; Statistical Learning Theory; Distributed Optimization; Learning under Data Heterogeneity and Distribution Shift; Sequential Anomaly Detection

UNDERGRADUATE RESEARCH

Optimal Client Selection in Federated Learning

2026 - Present

- B.Sc. research project investigating client selection under non-IID data and communication budgets using information-theoretic generalization bounds and KL-divergence-based distribution mismatch.
- Studying source-aware target-risk bounds, cost-aware sample allocation, and admission thresholds for excluding highly mismatched clients; reproducing supporting numerical experiments.

TEACHING EXPERIENCE

Teaching Assistant, Department of Electrical Engineering

Sep. 2024 - Present

Sharif University of Technology

Tehran, Iran

- Currently serve as a TA for Engineering Probability and Statistics; previously served as Head TA for Engineering Mathematics and as a TA for Linear Algebra, Signals and Systems, and Engineering Mathematics.
- Design and review exercises, quizzes, and assignments; lead structured problem-solving sessions and coordinate grading and recitation activities.
- Organized undergraduate Engineering Mathematics and Linear Algebra workshops through the Resana scientific society.

SELECTED COURSE PROJECTS

Sequence Anomaly Detection with OC-SVM, HMM/Viterbi, and HMA

- Implemented and evaluated OC-SVM baselines, Gaussian HMMs, log-space Viterbi decoding, and a simplified HMA pipeline on Wafer and synthetic Gaussian sequences.
- Achieved 0.9958 ROC-AUC with window-based OC-SVM on Wafer and 0.9907 F1 for Viterbi anomaly localization; analyzed the limitations of compact hidden-state representations.

Queueing Theory, Markov Chains, and Discrete-Event Simulation | *Python, LaTeX* Dec. 2024

- Modeled finite-capacity queueing systems with Poisson arrivals, exponential service times, feedback, and continuous-time Markov chains.
- Derived throughput and cost measures, then validated analytical results against reproducible discrete-event simulations and documented modeling assumptions in an English technical report.

License Plate Recognition under Blur | *Python, Jupyter, NumPy*

June 2025

- Built a closed-template recognition pipeline using grayscale preprocessing, thresholding, contour-based character segmentation, and correlation-based template matching.
- Estimated blur direction and applied spatial downsampling and first-order inverse filtering; packaged the workflow as a reproducible notebook with saved deblurred outputs.

TECHNICAL SKILLS

Programming and scientific computing: Python, NumPy, Matplotlib, MATLAB/Simulink, C++, Jupyter

Mathematical and computational methods: Probability and statistics, Markov chains, discrete-event simulation, linear algebra, differential equations, Fourier analysis, signal processing, convex optimization

Tools and languages: Git, LaTeX; Persian (native), English (reading and writing proficiency)